ALICE L WALTON SCHOOL OF MEDICINE

Sinus Bradycardia and Related Disorders

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Learning Objectives (5)

After completing this brick, you will be able to:

1. Explain the pathophysiology of sinus bradycardia and related disorders.
2. Describe the clinical findings associated with sinus bradycardia.
3. Explain the common causes of sinus bradycardia.
4. Outline a treatment plan for sinus bradycardia and related disorders.
5. Discuss escape rhythms as they relate to bradycardia.

CASE CONNECTION

RL, 68-year-old male presents to the emergency department with episodes of dizziness and mild confusion over the past week. On his ECG, a slow heart rate is noted. Medical student JD is on cardiology clerkship; they are asked to assess possible causes and determine management options.

What factors could be contributing to RL's bradycardia? What would be appropriate next steps in diagnostic and treatment considerations? Consider your answers as you read, and we'll revisit RL at the end of the brick.

GO TO CONCLUSION ↓

We have previously discussed the ECG-related features of Sinus Bradycardia. In this brick, we will discuss the pathophysiological origins of bradyarrhythmias and its consider treatment options.

What is sinus bradycardia?

The definition of sinus bradycardia is straightforward. It is a form of sinus rhythm in which the rate is abnormally slow, i.e. <60 bpm. As we will elaborate below, it can be clinically insignificant or cause a variety of clinical problems. In addition to pathology in the sinoatrial node (SAN) it can result from a variety of drugs and physiologic changes such as hypoxia and alteration in the output of the autonomic nervous system.

Figure 1 below demonstrates sinus bradycardia, rate 43 bpm.

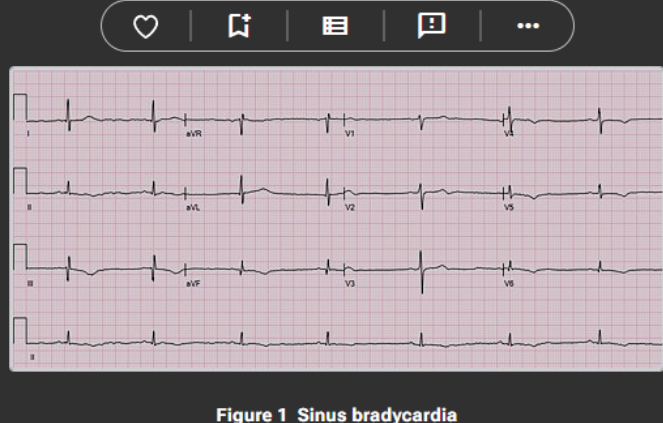

Figure 1 Sinus bradycardia

What are sinus bradycardia-related disorders?

Generally, Sick Sinus Syndrome (SSS) refers to the clinical spectrum of disorders related to sinus bradycardia. These disorders of rhythm are considered "close cousins" of sinus bradycardia and have in common the same pathophysiology as sinus bradycardia. To back up and take more of a "30,000 foot view", SSS is one of two broad categories of clinically significant bradycardia, the other being AV block. AV block is not the focus of this brick.

In addition to its defining disorder sinus bradycardia, SSS also includes sinus pause, chronotropic incompetence, and bradycardia-tachycardia syndrome. These are not mutually exclusive, often coexisting in the same patient. They reflect the varied expressions of sinoatrial node dysfunction. We've already discussed sinus bradycardia, so let's briefly consider the others.

Although mainly associated with bradycardia, in a subset of SSS, bradycardia alternates with atrial fibrillation, a condition known as "bradycardia-tachycardia syndrom

A sinus pause is an abrupt stoppage of SAN activity. Cardiac activity usually resumes after a variable amount of time which may range from just over a second to several seconds. In some instances cardiac activity may not resume at all and result in "flat line" or asystole, leading to syncope or death.

Figure 2 below illustrates an example of a sinus pause.

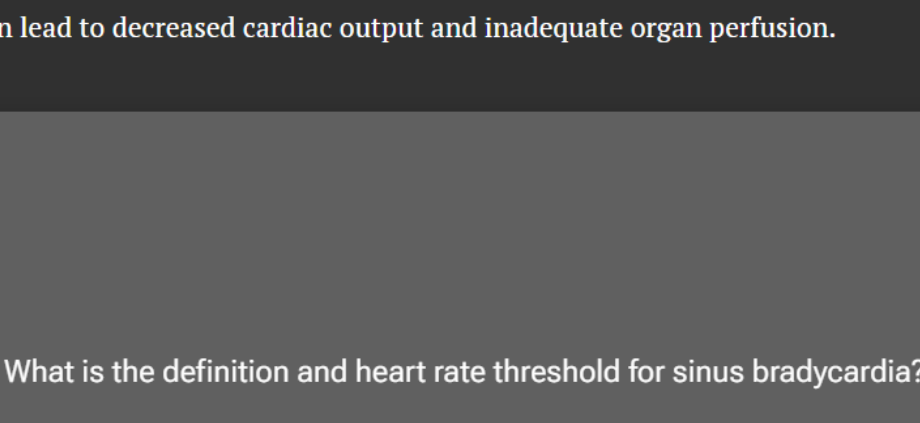

Figure 2 Sinus pause

What is the pathophysiology of sinus bradycardia? (LO#1)

Sinus bradycardia results from decreased automaticity of the sinoatrial (SA) node, leading to a heart rate below 60 beats per minute. This commonly occurs due to increased parasympathetic (vagal) tone, reduced sympathetic stimulation, or intrinsic dysfunction of SA nodal pacemaker cells. Slowed phase 4 depolarization in SA nodal cells decreases the frequency of action potential generation. If cardiac output becomes inadequate, compensatory mechanisms may fail, resulting in symptoms such as dizziness, fatigue, or syncope.

Cardiac Output Effects

Cardiac output equals heart rate multiplied by stroke volume, making bradycardia a potential threat to adequate tissue perfusion. When heart rate decreases, the body initially compensates by increasing stroke volume through enhanced ventricular filling (Frank-Starling mechanism) and increased contractility. However, this compensation has limits, and severe bradycardia can lead to decreased cardiac output and inadequate organ perfusion.

What is the definition and heart rate threshold for sinus bradycardia?

What Are the Common Clinical Presentations and Risk Factors Associated With Sinus Bradycardia? (LO#2)

The clinical presentation of sinus bradycardia varies significantly depending on the degree of brad and individual patient factors. Many ain asymptomatic, particularly young athletes or individuals with good cardiovascular fitness. However, when cardiac output becomes inadequate to meet tissue perfusion needs, patients develop characteristic symptoms that reflect decreased blood flow to vital organs, especially the brain and other metabolically active tissues.

Symptoms and Signs

The most **common symptoms** of clinically significant sinus bradycardia include dizziness, lightheadedness, and presyncope or syncope due to decreased cerebral perfusion. Patients may report feeling weak, tired, or having reduced exercise tolerance as cardiac output fails to increase appropriately with physical demands. Chest discomfort or shortness of breath can occur, particularly in patients with underlying coronary artery disease where reduced heart rate decreases coronary perfusion.

Physical examination findings in sinus bradycardia include a slow, regular pulse rate below 60 bpm. Blood pressure may be normal or low, depending on stroke volume compensation.

What Pathological Conditions and Diseases Commonly Cause Sinus Bradycardia? (LO#3)

Sinus bradycardia can result from a wide variety of pathological conditions affecting either the SA node directly or the autonomic nervous system's influence on heart rate. The causes can be broadly categorized into intrinsic cardiac abnormalities that directly affect the SA node (SAN), extrinsic systemic conditions that indirectly influence heart rate, and drug-induced bradycardia from therapeutic me

ACTIVE LEARNING QUESTION

A 75-year-old patient presents with dizziness and a heart rate of 45 bpm. Which risk factor is most likely contributing to his bradycardia?

- A. ○ Hyperthyroidism
- B. ● Age-related SA node degeneration
- C. ○ Cocaine use
- D. ○ Physical fitness
- E. ○ Hypokalemia

Explanation

The correct answer is Age-related SA node degeneration (B). Advanced age is associated with progressive fibrosis and calcification of the cardiac conduction system, particularly affecting the SA node. Hyperthyroidism (A) typically causes tachycardia, not bradycardia. Cocaine use (C) generally increases heart rate. Physical fitness (D) can cause physiological bradycardia but is less likely in a 75-year-old with symptoms. Hypokalemia (E) typically does not cause bradycardia.

Intrinsic factors Cardiac Pathology (intrinsic Cardiac Disease) Causing SAN Dysfunction

Intrinsic factors are inherent to cardiac pathology which can cause SAN dysfunction. SAN dysfunction is primarily a disease of the elderly and, in such patients, is most commonly caused by age related degeneration and fibrosis of the conduction system. Also, an increasingly recognized cause of conduction system disease in the elderly, including SAN dysfunction, is wild type transthyretin cardiac amyloidosis (ATTRwt).

CLINICAL CORRELATION

Transthyretin cardiac amyloidosis (ATTR) is a restrictive cardiomyopathy caused by deposition of misfolded transthyretin protein in the myocardium, leading to stiff ventricles and impaired diastolic filling. It may occur as a hereditary form due to TTR gene mutations or an age-related (wild type) form, commonly presenting with heart failure with preserved ejection fraction and conduction abnormalities. Conduction abnormalities such as sinus bradycardia are often seen in cardiac amyloidosis.

In younger patients, SAN dysfunction is more likely to be due to congenital heart disease or autonomic nervous system imbalance, known as vasovagal syndrome, to be discussed below.

Extrinsic (Extracar

Extrinsic factors influence SA node function through systemic effects rather than direct cardiac involvement. Hypothyroidism significantly slows heart rate by decreasing metabolic rate and reducing the heart's responsiveness to sympathetic stimulation. Electrolyte imbalances, particularly hyperkalemia, can suppress SA node automaticity and slow conduction through the myocardium.

Increased intracranial pressure from any cause can stimulate the vagus nerve, leading to bradycardia as part of the Cushing reflex. Hypothermia slows all metabolic processes, including SA node automaticity.

An important extracardiac influence on SAN activity is the vagus nerve. Vagal activity suppresses SAN activity and thereby slows heart rate. This can occur in physiologic form with athletic conditioning. A pathologic form of excessive vagal activity can cause symptomatic bradycardia and pauses (as well as AV block). It is known as vasovagal syndrome. It can occur at all ages, and its episodes are generally circumstantial (eg micturition syncope, passing out at the sight of blood, carotid sinus syncope, and some instances of postural syncope and dizziness).

Drug-Induced Bradycardia

Medications represent a common and reversible cause of sinus bradycardia. Beta-blockers reduce heart rate by blocking sympathetic stimulation of beta-1 receptors in the SA node. Non-dihydropyridine calcium blockers like verapamil and diltiazem directly suppress SA node automaticity by blocking calcium channels essential for pacemaker cell function.

Digoxin increases va node function. Other medications including clonidine and certain antidepressants can also cause bradycardia through various mechanisms affecting autonomic function or direct cardiac effects.

What Is the Treatment for Sinus Bradycardia and Sick Sinus Syndrome? (LO#4)

Diagnostic Approach

The diagnostic evaluation begins with a detailed history focusing on symptom onset, duration, and triggers, along with a comprehensive medication review. Physical examination should assess vital signs, cardiovascular status, and signs of underlying systemic disease. A 12-lead electrocardiogram (ECG) provides essential information about heart rate, rhythm, and any associated conduction abnormalities.

Laboratory studies should include thyroid function tests, electrolytes (particularly potassium and calcium), and cardiac biomarkers if ischemia is suspected. Ambulatory ECG monitoring, either through Holter monitoring or event recorders, can capture intermittent bradycardia and assess heart rate variability. It can also establish correlation between bradycardic episodes and symptoms. Exercise stress testing may reveal chronotropic incompetence, which was discussed above.

Differential Diagnosis

Distinguishing sick sinus syndrome from other causes of bradycardia requires careful analysis of clinical presentation and diagnostic testing. Medication-induced bradycardia ship with drug initiation and resolves ation. Hypothyroidism-related bradycardia improves with thyroid hormone replacement and is associated with other signs of hypothyroidism.

Athletic bradycardia occurs in physically fit individuals who are typically asymptomatic and demonstrate appropriate heart rate response to exercise. Vagally-mediated bradycardia often occurs in specific situations or with particular triggers and may be associated with vasovagal symptoms. Ischemic causes may have associated chest pain, ECG changes, or elevated cardiac biomarkers.

ACTIVE LEARNING QUESTION

Which finding is most characteristic of sick sinus syndrome

- A. ○ Associated hypothyroidism
- B. ● Sinus pauses greater than 3 seconds
- C. ○ Beta-blocker use
- D. ○ First-degree AV block

Explanation

The correct answer is Sinus pauses greater than 3 seconds (B). Prolonged sinus pauses are a hallmark feature of sick sinus syndrome, indicating sig roidism, bradycardia is typically persistent and attributable to reduced metabolic demand, rather than to intermittent sinus node dysfunction. Beta-blocker use (C) is a reversible, extrinsic cause. First-degree AV block (D) involves the AV node, not the SA node.

Treatment of sinus bradycardia depends on the underlying cause, severity of symptoms, and hemodynamic stability. The primary goals are to restore adequate heart rate and cardiac output, alleviate symptoms, and prevent complications such as syncope or falls. Management strategies range from observation and lifestyle modifications in asymptomatic patients to emergency interventions and permanent pacemaker implantation in severe cases. The approach must be individualized based on patient factors, comorbidities, and the reversibility of the underlying cause.

Indications for Treatment

Treatment is indicated when sinus bradycardia causes symptoms or hemodynamic compromise and is intrinsic to the patient and not expected to resolve. Symptomatic bradycardia with dizziness, syncope, presyncope, fatigue, or exercise intolerance requires intervention.

Asymptomatic bradycardia in athletes or young, healthy individuals typically requires only observation and reassurance. However, even asymptomatic patients may need treatment if they have very slow heart rates (typically below 40 bpm), prolonged pauses, or chronotropic incompetence that could predispose to future

Treatment

Acute management of symptomatic bradycardia includes atropine as the first-line medication, which blocks vagal stimulation and increases heart rate. It acts by competitively blocking muscarinic (M_2) receptors in the heart, thereby inhibiting parasympathetic (vagal) tone at the sinoatrial (SA) node. Atropine is most effective for bradycardia caused by increased vagal tone but may be less effective in intrinsic SA node disease. The most common adverse effects of atropine include tachyarrhythmia, constipation, xerostomia, blurred vision, and photophobia.

If atropine is ineffective, dopamine (D1 and D2 agonist) or epinephrine (α and β agonist) infusions can provide chronotropic support while preparing for more definitive treatment.

Also, when atropine is ineffective, **transcutaneous pacing** may provide temporary heart rate support in emergency situations and serve as a bridge to invasive (temporary transvenous) pacing if needed.

Transcutaneous pacing (TCP) is a temporary, noninvasive method of cardiac pacing used to treat symptomatic bradycardia or high-degree atrioventricular (AV) block when the heart rate is too slow to maintain adequate perfusion. Large pacing pads placed on the chest deliver electrical impulses through the skin to stimulate ventricular depolarization and contraction. TCP is typically used as an emergency or bridge therapy while definitive treatment (e.g., medications, transvenous pacing) is arranged.

CLINICAL CORREL

Transvenous pacing is an invasive, temporary pacing technique in which a pacing catheter is inserted through a central vein and positioned in the right ventricle to deliver electrical impulses directly to the myocardium. It is used for persistent or unstable bradyarrhythmias when medications or transcutaneous pacing are ineffective or not tolerated.

However, transcutaneous pacing is difficult to implement and lacks good evidence for improved patient outcomes.

The primary long-term treatment for symptomatic sick sinus syndrome and other forms of intrinsic SA node dysfunction is permanent pacemaker implantation. Modern pacemakers can provide rate-responsive pacing that adjusts heart rate based on physiological demands.

In considering treatment, two other situations deserve discussion: atrial fibrillation as part of brady-tachy syndrome and vasovagal syndrome.

In brady-tachy syndrome patients may need a combination of drug treatment (to control or suppress the atrial fibrillation) and a pacemaker for the bradycardia.(brady-tachy syndrome is beyond the discussion for this brick)

Vasovagal episodes can also manifest as a combination of bradycardia (the cardio inhibitor component) and hypotension (the vasodepressor component). Therefore, management is difficult and must be directed to both the bradycardia and the hypotension. Management is individualized, as one component may predominate over the other in any given patient.

Treatment starts with lifestyle interventions including maintenance of hydration and situati n cite known triggers. Medications to improve the hypotension component (midodrine, fludrocortisone) may be helpful. Pacing is employed as a last resort but is not indicated in most patients.

What Are Escape Rhythms? (LO#5)

Fortunately, stoppage of the sinoatrial node and AV block do not usually lead to cardiac arrest. In many instances, pacemakers below the level of the sinoatrial node (SAN) or the block take over, producing what are known as **escape rhythms.** Pacemaker cells exist at multiple levels of the cardiac conduction system—including the AV node, bundle of His, bundle branches, and ventricular myocardium—and act as subsidiary (rescue) pacemakers, taking over by default when impulses from higher centers fail to arrive in a timely manner.

When SAN activity is severely slowed or absent, or in complete AV block, the fastest available pacemaker in the conduction system assumes control. Pacemakers located lower in the conduction system fire at slower intrinsic rates, so escape rhythms following SAN failure are typically faster than those seen with AV block. In severe sinus bradycardia or sinus arrest, the AV node becomes the dominant pacemaker, producing a **junctional (AV nodal) escape rhythm.** Although escape rhythms are protective and usually require no immediate treatment, they are often unreliable; therefore, the underlying SAN dysfunction generally requires permanent pacemaker therapy.

The clinical manifestations of a junctional escape rhythm are essentially those of the primary disturbance, that is, SAN dysfunction. Given that the typical discharge rate of an AVN escape pacemaker is around 40-60 bpm, the patient may be asymptomatic

On an ECG, the P waves are typically absent although sometimes they may be inverted and appear before or after the QRS. QRS complexes are narrow because the impulses are conducted down the bilateral bundle branches.

Figure 3 below demonstrates a junctional escape rhythm.

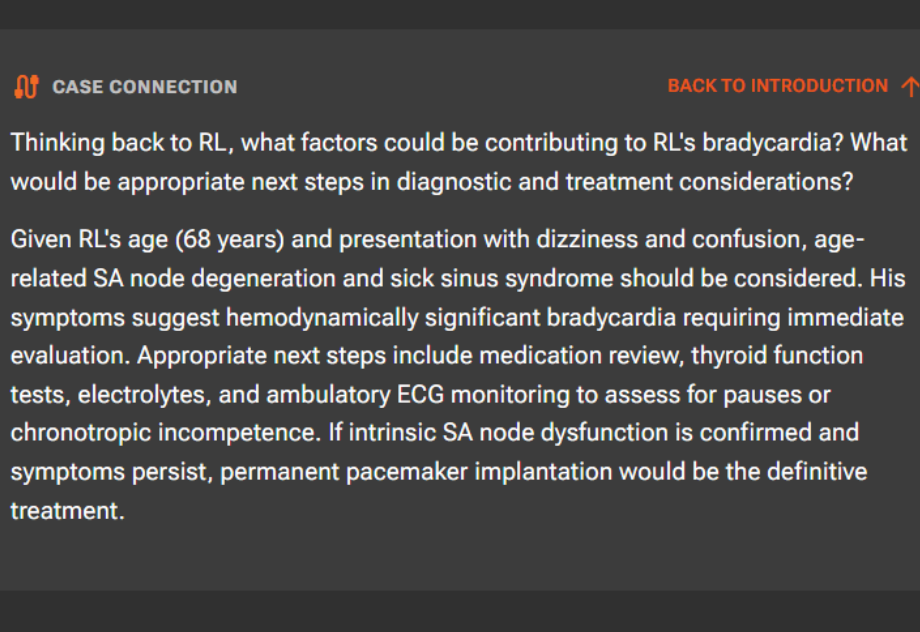

Figure 3 Junctional escape rhythm

CLINICAL CORRELATION

Less common varian unctional rhythm, in which the rate is 60 t dia, in which the rate is more than 100 beats/min. These are not purely escape rhythms and sometimes usurp the dominant cardiac rhythm rather than occurring by default, as would a true escape rhythm. It deserves mention here but is beyond the scope of this brick.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to RL, what factors could be contributing to RL's bradycardia? What would be appropriate next steps in diagnostic and treatment considerations?

Given RL's age (68 years) and presentation with dizziness and confusion, age-related SA node degeneration and sick sinus syndrome should be considered. His symptoms suggest hemodynamically significant bradycardia requiring immediate evaluation. Appropriate next steps include medication review, thyroid function tests, electrolytes, and ambulatory ECG monitoring to assess for pauses or chronotropic incompetence. If intrinsic SA node dysfunction is confirmed and symptoms persist, permanent pacemaker implantation would be the definitive treatment.

Summary

- Sinus bradycardia is defined as a heart rate below 60 bpm originating from the SA node, which can significantly impact cardiac output when compensation mechanisms fail.
- Common presentations include dizziness, syncope, fatigue, and exercise intolerance, with iac disease being most susceptible
- Major risk factors include advanced age, medications (beta-blockers, calcium channel blockers, digoxin), hypothyroidism, and intrinsic cardiac diseases affecting the SA node.
- Pathological causes are categorized as intrinsic cardiac (ischemia, infiltrative diseases, degeneration) or extrinsic factors (hypothyroidism, medications, electrolyte imbalances, increased vagal activity).
- Sick sinus syndrome is characterized by intrinsic SA node dysfunction with sinus pauses greater than 3 seconds, chronotropic incompetence, and alternating bradycardia-tachycardia.
- Treatment depends on symptoms and hemodynamic stability, ranging from observation in asymptomatic patients to permanent pacemaker implantation for sick sinus syndrome.
- Acute management includes atropine for vagally-mediated bradycardia, while permanent dual-chamber pacing with rate-responsive features is the definitive treatment for symptomatic intrinsic SA node dysfunction.
- Escape rhythms occur when a subsidiary pacemaker takes over due to failure or suppression of the primary pacemaker, resulting in a slower, protective rhythm that maintains cardiac output.

Review Questions

1. What defines sinus bradycardia on an ECG?

- A. ○ Heart rate above 100 bpm
- B. ● Heart rate under 60 bpm
- C. ○ Irregular
- D. ○ Pauses longer than 3 seconds
- E. ○ No visible P waves

Hide Explanation

The correct answer is (B) a heart rate below 60 bpm with normal sinus rhythm. Sinus bradycardia is specifically defined by a heart rate under 60 bpm while maintaining normal atrial and ventricular activation originating from the SA node. (A) Elevated heart rate describes tachycardia, not bradycardia. (C) Irregular rhythm is not a feature of sinus bradycardia. (D) Pauses longer than 3 seconds indicate sick sinus syndrome, not sinus bradycardia alone. (E) Absence of P waves suggests other arrhythmias, not sinus bradycardia.

2. Which symptom is most commonly associated with sinus bradycardia?

- A. ● Dizziness and lightheadedness
- B. ○ Chest discomfort
- C. ○ Palpitations
- D. ○ Fever
- E. ○ High blood pressure

Hide Explanation

The correct answer is (A) dizziness. Dizziness, lightheadedness, and presyncope are the typical symptoms caused by reduced cerebral perfusion in bradycardia. (B) Chest discomfort occurs but is more linked with coronary artery disease and not the most common symptom. (C) Palpitations are more associated with tachyarrhythmias, not bradycardia. (D) Fever is unrelated to bradycardia. (E) Hypertension is not a typical presentation of sinus bradycardia.

3. Which condition commonly causes intrinsic SA node dysfunction leading to bradycardia?

- A. ○ Hypothyroidism
- B. ○ Hyperkalemia
- C. ● Ischemic heart disease
- D. ○ Vasovagal response
- E. ○ Athletic conditioning

Hide Explanation

The correct answer is (C) ischemic heart disease. Ischemia, especially involving the right coronary artery, directly impairs SA node function. (A) Hypothyroidism causes extrinsic bradycardia intrinsic SA node disease. (B) Hyperkalemia is an extrinsic factor affecting conduction but not intrinsic SA node structure. (D) Vasovagal response increases vagal tone, an extrinsic influence. (E) Athletic conditioning causes physiological, not pathological, bradycardia.

4. What is the definitive long-term treatment for symptomatic sick sinus syndrome?

- A. ○ Long-term atropine therapy
- B. ○ Beta-blocker therapy
- C. ○ Digoxin therapy
- D. ● Permanent pacemaker implantation
- E. ○ Observation only

Hide Explanation

The correct answer is (D) permanent pacemaker implantation. This is the gold standard treatment for symptomatic intrinsic SA node dysfunction, restoring appropriate heart rate. (A) Long-term atropine is impractical and ineffective long-term. (B) Beta-blockers worsen bradycardia and are contraindicated. (C) Digoxin suppresses SA node function and is inappropriate. (E) Observation is unsafe in symptomatic patients needing intervention.

5. Which of the following is true of escape rhythms in sick sinus syndrome?

- A. ○ They usurp the normal rhythm
- B. ● They take over the rhythm by default because the sinus mechanism is deficient.
- C. ○ They should be suppressed by antiarrhythmic drugs
- D. ○ They usually arise from one of the bundle branches
- E. ○ They have a wide QRS complex

Hide Explanation

The correct answer is (B). During normal sinus rhythm the SAN fires at a rate exceeding those of the lower pacemakers, depolarizing them with each cardiac cycle before their action potentials have opportunity to fire spontaneously. However, if SAN activity ceases or the sinus rate falls below that of the next fastest pacemaker, that pacemaker will discharge by default, resulting in the escape rhythm. Although lower pacemakers can usurp the cardiac rhythm (A), when they do, they are known as accelerated, not escape, rhythms. Escape rhythms may serve a useful function, providing a pacemaker to mitigate symptoms and in some cases preserve life. To suppress them would be harmful (C). Treatment is directed toward the primary disturbance, which is the SAN dysfunction. The escape rhythm is not the primary disturbance. When the SAN fails, the fastest available pacemaker takes over the rhythm, which is located in the AV node, part of the AV junction (D). Because the escape rhythm originates in the AV junction, the impulse will travel down the bilateral bundle bran

6. A 69-year-old man presents to the emergency department with dizziness and lightheadedness. His heart rate is 38 bpm and blood pressure is 92/56 mmHg. ECG shows sinus bradycardia. He recently experienced a vasovagal episode while having blood drawn. There is no history of intrinsic conduction system disease. Which of the following best explains why atropine is an appropriate initial treatment in this patient?

- A. ○ Atropine directly increases ventricular contractility
- B. ● Atropine blocks parasympathetic input to the sinoatrial node
- C. ○ Atropine restores sinoatrial node automaticity in fibrotic tissue
- D. ○ Atropine provides definitive long-term treatment for bradyarrhythmia

Hide Explanation

The correct answer is Atropine blocks parasympathetic input to the sinoatrial node (B). Atropine is a muscarinic (M_2) receptor antagonist that inhibits parasympathetic (vagal) tone at the sinoatrial node. In this patient, the bradycardia is caused by increased vagal activity during a vasovagal episode, making atropine an effective first-line treatment to increase heart rate and improve perfusion. Atropine does not increase myocardial contractility (A). Its primary action is on heart rate through vagolytic effects at the SA and AV nodes. Increased contractility is mediated by beta-adrenergic stimulation, not muscarinic blockade. Atropine is ineffective in intrinsic SA node disease (C), such as age-related fibrosis or sick sinus syndrome. In those cases, the underlying ve treatment is permanent pacemaker implantation. Atropine is a short-term, acute intervention (D). It is not suitable or safe for long-term management of symptomatic bradycardia. Persistent or intrinsic causes require alternative therapies such as pacing.

How would you rate this Brick?

References

- [redacted]
- [redacted] *Marriott's Practical Electrocardiography* [redacted] *ProQuest Ebook Central* [redacted]

[redacted]

- [redacted] *Marriott's Practical Electrocardiography* [redacted] *ProQuest Ebook Central* [redacted]

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

- 01 ☑ ~~Explain the pathophysiology of sinus bradycardia and related disorders.~~
- 02 ☑ ~~Describe the clinical findings associated with sinus bradycardia.~~
- 03 ☑ ~~Explain the common causes of sinus bradycardia.~~
- 04 ☑ ~~Outline a treatment plan for sinus bradycardia and related disorders.~~
- 05 ☑ ~~Discuss escape rhythms as they relate to bradycardia.~~

Objective Confidence: 5/5